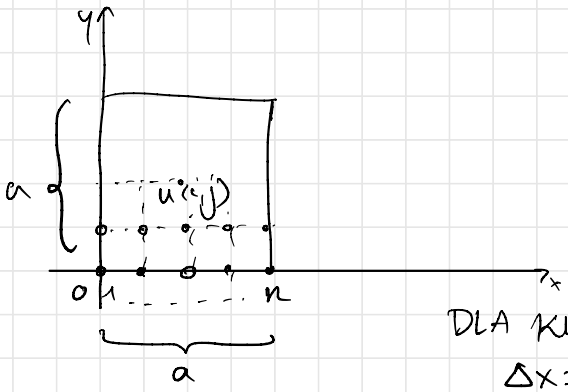
Metody numeryczne:

I. Metoda relaksacyjna



DLA KWADRATU

$$\Delta x = \frac{a}{N}$$

$$\Delta y = \frac{a}{N} = \Delta x$$

$$\frac{\partial v}{\partial x} = \lim_{\Delta x \to 0} \frac{v(x+\Delta x, y) - v(x,y)}{\Delta x}$$

$$\frac{\partial u}{\partial x} \approx \frac{u(x+\Delta x, y) - v(x,y)}{\Delta x}$$

$$h = \Delta x = \Delta x$$

$$x_i = \Delta x \cdot i$$

$$y_j = \Delta x \cdot j = \Delta y \cdot j$$

$$u(x,y) \longrightarrow u_{i,j}$$

$$u(x+h, y) \longrightarrow u_{i+1,j}$$

$$u(x, y+h) \longrightarrow u_{i,j+1}$$

$$\frac{\partial u}{\partial x}(x_i, y_j) \approx \frac{u_{i+1,j} - u_{i,j}}{h}$$

$$\frac{\partial v}{\partial y}(x_i, y_j) \approx \frac{u_{i,j+1} - u_{i,j}}{h}$$

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{\frac{\partial u}{\partial x}(x_{i+1}, y_j) - \frac{\partial u}{\partial x}(i,j)}{h}$$

$$= \frac{\frac{u(x_{i+2}, y_j) - u(x_{i+1}, y_i)}{h} - \frac{u(x_{i+1}, y_i) - u(x_i, y_j)}{h}}{h}$$

$$= \frac{u(x_{i+2}, y_j) - 2u(x_{i+1}, y_i) + u(x_i, y_j)}{h^2}$$

$i \to i-1$

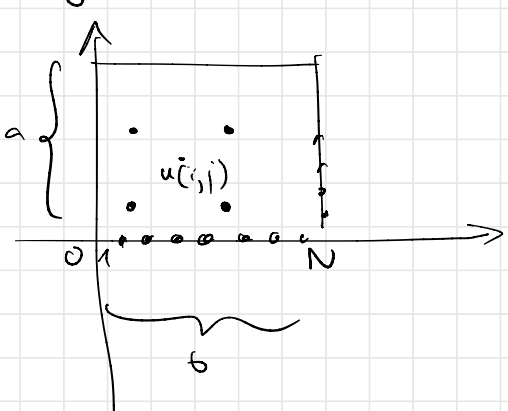
$$\frac{\partial^2 u}{\partial x^2}(x_i, y_j) \approx \frac{u(x_{i+1}, y_j) - 2u(x_i, y_j) + u(x_{i-1}, j)}{h^2}$$

$$\frac{\partial^2 u}{\partial y^2}(x_i, y_j) = \frac{u(x_i, y_{j+1}) - 2u(x_i, y_j) + u(x_i, y_{j-1})}{h^2}$$

$+$

$$\frac{1}{h^2}\left(u(x_{i+1}, y_j) + u(x_{i-1}, y_j) + u(x_i, j+1) + u(x_i, y_{j-1}) - 4u(x_i, y_i)\right) = 0 \quad | \cdot h^2$$

$$u(x_i, y_i) = \frac{u(x_{i+1}, y_j) + u(x_{i-1}, y_j) + u(x_i, y_{j+1}) + u(x_i, y_{j-1})}{4}$$



$u(0,j) = f(j)$

$u(N,j) = f(j)$

$u(i,0) = f(i)$

$u(i,N) = f(i)$

WYKŁAD
09.03.2026

Metody rozwiązywania równań Laplace'a

↓

Rozdzielanie zmiennych

weźmy równanie Laplace'a 2D:

1) $\partial_x^2 V + \partial_y^2 V = 0 \qquad (x,y) \in \Omega$

2) $V(x,y)\Big|_{\partial\Omega} = f(x,y)$ - dane

załóżmy $V(x,y) = X(x) \cdot Y(y) \longrightarrow \nabla^2 [X(x) \cdot Y(y)] = 0$

$$X''(x) \cdot Y(y) + Y''(y) \cdot X(x) = 0 \quad / : (Y \cdot X)$$

$$\frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)}$$

zależy od x

ponieważ lewa strona równania $[LHS(x)]$, a prawa strona równania $[RHS(y)]$ zależy od y to $x \leftrightarrow y$ są niezależne

$$\frac{X''(x)}{X(x)} = \text{const} = -\frac{Y''(y)}{Y(y)}$$

$$\text{const} = -k \qquad k > 0 \quad k \in \mathbb{R}$$

$$\begin{cases} X''(x) = -k \cdot X(x) \\ Y''(y) = k \cdot Y(y) \end{cases}$$

równania różniczkowe zwyczajne

$$X(x) = A \cdot \cos(\omega \cdot x) + B \cdot \sin(\omega \cdot x)$$

$$\omega = \sqrt{k}$$

$Y(y) = ?$ $\quad Y''(y) = \omega^2 Y(y)$

$$Y(y) = Ce^{+\omega y} + De^{-\omega y}$$

A, B, C, D - cztery dowolne stałe

DRGANIA:
DYGRESJA

$m x''(t) = -k x(t) \quad x'' = -\omega^2 x$

$\omega = \sqrt{\frac{k}{m}} \quad \left[\frac{1}{s}\right]$

$x(t) = \tilde{A}\cos(\omega t + \varphi)$

osc. harm:

$x(t) = A\cos(\omega t) + B(\sin(\omega t))$

$$V(x,y) = \left(A\cos(\omega x) + B\sin(\omega x)\right)\left(Ce^{\omega y} + De^{-\omega y}\right)$$

chcemy dopasować do warunku brzegowego

$$V(x,y)\Big|_{\partial\Omega} = f(x,y)$$

równanie Laplace'a jest równaniem liniowym

że jeśli V_1, V_2 - rozwiązania równania Laplace'a

$\alpha_1 \cdot V_1 + \alpha_2 \cdot V_2 = V_3$ też jest rozwiązaniem

$$\partial_x^2(\alpha_1 \cdot V_1 + \alpha_2 V_2) = \alpha_1 \partial_x^2 V_1 + \alpha_2 \partial_x^2 V_2$$

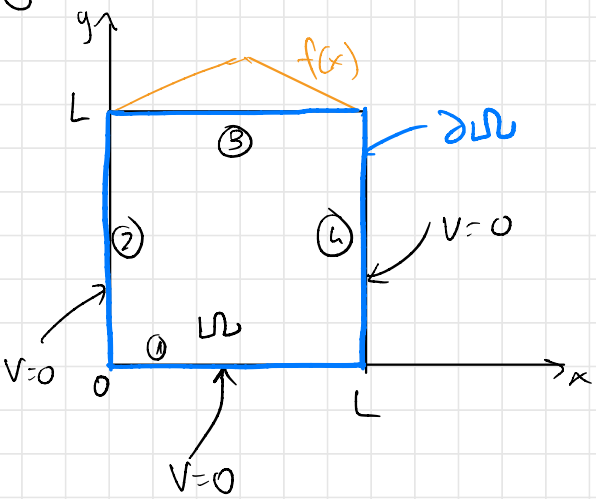
Możemy założyć, że ogólne rozwiązanie równania Laplace'a

$$V(x,y) = \sum_{n=1}^{\infty} \left(A_n \cos(\omega x) + B_n \sin(\omega x)\right)\left(C_n e^{\omega y} + D_n e^{-\omega y}\right)$$

A_n, B_n, C_n, D_n – nieskończenie wiele stałych

wyznaczamy je z warunków brzegowych (boundary condition)

Przykład



$$\Omega = [0,L] \times [0,L] \ni (x,y)$$

$$\partial\Omega = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3 \cup \Gamma_4$$

dla $(x,y) \in \Gamma_1 \quad v(x,0) = 0$

$$\sum_{n=1}^{\infty} \left(A_n \cos(\omega x) + B_n \sin(\omega x)\right)(C_n + D_n) = 0$$

x - dowolne $\Rightarrow \forall n: C_n + D_n = 0$ (część niezależna od x)

$C_n = -D_n$

dla $(x,y) \in \Gamma_2$: $V(0,y) = 0$

$$\sum_{n=1}^{\infty} (A_n \cdot 1 + 0)\, C_n \left(e^{xy} - e^{\bar{x}y}\right) = 0$$

y - dowolne $\Rightarrow \forall n$: $C_n \neq 0 \quad A_n = 0$

$\Gamma_3 \ni (x,y)$: $V(L,y) = 0$

$$\sum_{n=1}^{\infty} B_n \sin(\omega L) \cdot C_n \left(e^{\omega y} - e^{-\omega y}\right) = 0$$

y - dowolne $\Rightarrow \forall n$: ~~$B_n \neq 0$~~ ?? bez sensu (rozwiązanie zerowe)

zakładamy: $B_n \neq 0$: $\sin(\omega L) = 0$

$$\omega L = \pi n, \quad n \in \mathbb{N}$$
$$\hookrightarrow \omega = \frac{n\pi}{L}$$

Na tę chwilę:

$$V(x,y) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\cdot\pi}{L} x\right) C_n \left(e^{\frac{n\pi}{L} y} - e^{-\frac{n\pi}{L} y}\right)$$

oznaczamy: $B_n \cdot C_n = \tilde{A}_n$

$$V(x,y) = \sum_{n=1}^{\infty} \tilde{A}_n \sin\left(\frac{n\pi}{L} x\right)\left(e^{\frac{\pi n}{L} y} - e^{-\frac{n\pi}{L} y}\right)$$

Dla $(x,y) \in \Gamma_4$: $V(x,L) = f(x)$

$$V(x,L) = \sum_{n=1}^{\infty} \tilde{A}_n \sin\left(\frac{n\pi}{L} x\right)\left(e^{n\pi} - e^{-n\pi}\right) = f(x) \quad \Big| \cdot \sin\left(\frac{m\pi}{L} x\right)$$

warunek ortogonalności $\rightarrow$

$$\int_0^L \sin\left(\frac{n\pi}{L} x\right) \sin\left(\frac{m\pi}{L} \cdot x\right) dx = \begin{cases} 0 : n \neq m \\ \frac{L}{2} : n = m \end{cases}$$

$\downarrow$

wektory bazowe

$\hat{e}_1, \hat{e}_2, \hat{e}_3 \longrightarrow \hat{e}_1 \circ \hat{e}_2 = 0 \quad \hat{e}_1 \circ \hat{e}_1 = 1 \rightarrow \hat{e}_n \circ \hat{e}_m = \delta_{nm}$

$$\int_0^L \sum_{n=1}^{\infty} \tilde{A}_n \sin\left(\frac{n\pi}{L}x\right)\sin\left(\frac{m\pi}{L}x\right)\left(e^{n\pi}-e^{-n\pi}\right)dx = \int_0^L f(x)\sin\left(\frac{m\pi}{L}x\right)dx \Big| \int_0^L dx$$

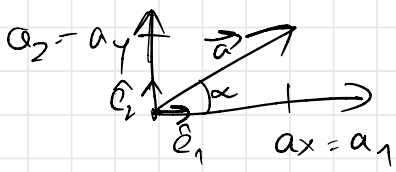
$\tilde{A}_n \cdot \frac{L}{2}$

$n = m$:

$$\tilde{A}_m\left(e^{m\pi}-e^{-m\pi}\right)\frac{L}{2} = \int_0^L f(x)\sin\left(\frac{m\pi}{L}x\right)dx$$

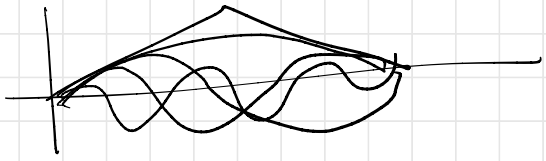
$$\tilde{A}_m = \frac{2}{L\left(e^{m\pi}-e^{-m\pi}\right)} \int_0^L f(x)\sin\left(\frac{m\pi}{L}x\right)dx$$

Wieh $\vec{a}$ -weletor $a_i = ?$



$$a_i = \vec{a} \circ \hat{e}_i$$

$\tilde{A}_m$:



$$\tilde{A}_m = \frac{4}{m^2\pi^2} \quad : \quad m = 1,3,5,\dots$$

$$\tilde{A}_n = 0 \quad : \quad m = 2,4,6,\dots$$